In [3]:
```
#Setup and Imports

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore', category=FutureWarning)
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from sklearn.metrics import (accuracy_score, precision_score, recall_scor
                             f1_score, roc_auc_score, confusion_matrix,
                             classification_report, roc_curve)
from sklearn.impute import SimpleImputer

# Set style for better visualizations
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_palette("husl")

print("="*60)
print("FINANCIAL INCLUSION PREDICTION SYSTEM")
print("Global Findex Database 2025 Analysis")
print("="*60)
```

```
============================================================
FINANCIAL INCLUSION PREDICTION SYSTEM
Global Findex Database 2025 Analysis
============================================================
```

In [4]:
```
# Data Loading

DATA_PATH = 'global_findex_2025.csv'

df = pd.read_csv(DATA_PATH, low_memory=False)

print(f"Dataset Shape: {df.shape[0]:,} rows × {df.shape[1]} columns")
print(f"\nFirst 3 rows preview:")
df.head(3)
```

```
Dataset Shape: 144,090 rows × 199 columns

First 3 rows preview:
```

Out[4]:

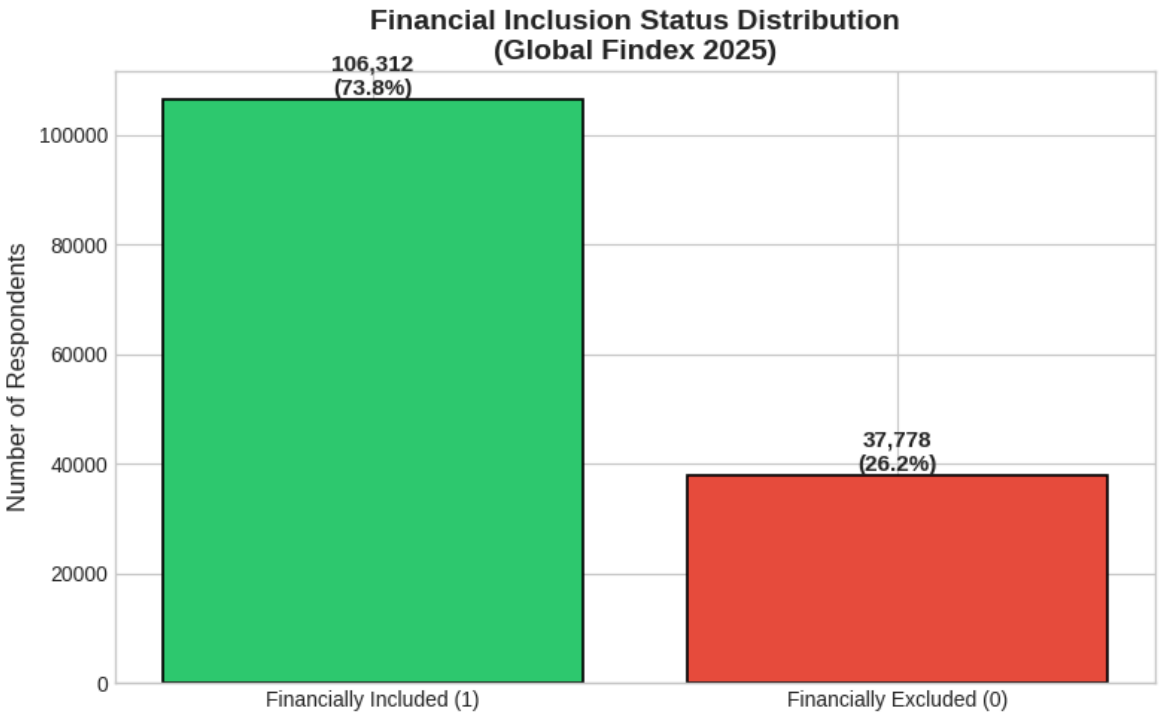
| | year | economy | economycode | regionwb | pop_adult | wpid_random | wgt | fem |
|---|---|---|---|---|---|---|---|---|
| **0** | 2024 | Nicaragua | NIC | Latin America & Caribbean (excluding high income) | 4830854 | 111111345 | 0.927315 | |
| **1** | 2024 | Costa Rica | CRI | Latin America & Caribbean (excluding high income) | 4118576 | 111111579 | 1.383884 | |
| **2** | 2024 | Mali | MLI | Sub-Saharan Africa (excluding high income) | 12725942 | 111112511 | 1.323486 | |

3 rows × 199 columns

In [5]:

```
# Target Variable Analysis

# Target variable distribution
print("TARGET VARIABLE DISTRIBUTION (account ownership)")
print("-" * 40)
target_counts = df['account'].value_counts()
target_pct = df['account'].value_counts(normalize=True) * 100

for val, count, pct in zip(target_counts.index, target_counts.values, tar
    label = "Has Account (Included)" if val == 1 else "No Account (Exclud
    print(f"{label}: {count:,} ({pct:.1f}%)")

# Visual
fig, ax = plt.subplots(figsize=(8, 5))
colors = ['#2ecc71', '#e74c3c']
labels = ['Financially Included (1)', 'Financially Excluded (0)']
ax.bar(labels, target_counts.values, color=colors, edgecolor='black', lin
ax.set_ylabel('Number of Respondents', fontsize=12)
ax.set_title('Financial Inclusion Status Distribution\n(Global Findex 202
for i, (v, p) in enumerate(zip(target_counts.values, target_pct.values)):
    ax.text(i, v + 1000, f'{v:,}\n({p:.1f}%)', ha='center', fontsize=11,
plt.tight_layout()
plt.savefig('01_target_distribution.png', dpi=150, bbox_inches='tight')
plt.show()
```

```
TARGET VARIABLE DISTRIBUTION (account ownership)
----------------------------------------
Has Account (Included): 106,312 (73.8%)
No Account (Excluded): 37,778 (26.2%)
```




In [8]:
```
# Feature Selection for Modeling
# Select key features based on problem definition

# Define feature categories
FEATURE_DICT = {
    'Demographics': ['female', 'age', 'educ', 'inc_q', 'emp_in', 'urbanic
    'Digital Connectivity': ['internet_use'],
    'Financial Behavior': ['saved', 'borrowed', 'receive_wages', 'receive
                           'receive_pensions', 'anydigpayment', 'pay_utili
                           'domestic_remittances', 'merchantpay_dig']
}

# Create feature list
selected_features = []
for category, features in FEATURE_DICT.items():
    selected_features.extend(features)
    # Check which features exist in data
    existing = [f for f in features if f in df.columns]
    missing = [f for f in features if f not in df.columns]
    if missing:
        print(f"  {category}: Missing features - {missing}")

# Final feature list (only existing columns)
features = [f for f in selected_features if f in df.columns]
target = 'account'

print(f"\n✓ Final Feature Count: {len(features)}")
print(f"✓ Target Variable: {target}")
print("\nFeatures by Category:")
for cat, feats in FEATURE_DICT.items():
    existing = [f for f in feats if f in df.columns]
    print(f"  {cat}: {existing}")
```

```
✓ Final Feature Count: 16
✓ Target Variable: account

Features by Category:
  Demographics: ['female', 'age', 'educ', 'inc_q', 'emp_in', 'urbanicity']
  Digital Connectivity: ['internet_use']
  Financial Behavior: ['saved', 'borrowed', 'receive_wages', 'receive_tran
sfers', 'receive_pensions', 'anydigpayment', 'pay_utilities', 'domestic_re
mittances', 'merchantpay_dig']
```

In [9]:

```
# Data Preprocessing
# Preprocessing pipeline

print("="*60)
print("DATA PREPROCESSING PIPELINE")
print("="*60)

# Create working copy
df_processed = df[features + [target]].copy()

# Step 1: Handle Missing Values
print("\n[1] MISSING VALUE ANALYSIS")
print("-" * 40)
missing_before = df_processed.isnull().sum()
missing_pct = (missing_before / len(df_processed) * 100).round(2)
missing_summary = pd.DataFrame({
    'Missing Count': missing_before,
    'Missing %': missing_pct
})
missing_summary = missing_summary[missing_summary['Missing Count'] > 0].s
print(missing_summary.head(10))

# Impute missing values
# Numeric: median, Categorical: mode
numeric_cols = df_processed.select_dtypes(include=[np.number]).columns.to
categorical_cols = df_processed.select_dtypes(include=['object']).columns

if len(numeric_cols) > 0:
    num_imputer = SimpleImputer(strategy='median')
    df_processed[numeric_cols] = num_imputer.fit_transform(df_processed[n

if len(categorical_cols) > 0:
    cat_imputer = SimpleImputer(strategy='most_frequent')
    df_processed[categorical_cols] = cat_imputer.fit_transform(df_process

print(f"\n✓ Missing values after imputation: {df_processed.isnull().sum()

# Step 2: Encode Categorical Variables
print("\n[2] ENCODING CATEGORICAL VARIABLES")
print("-" * 40)
label_encoders = {}
for col in categorical_cols:
    if col != target:
        le = LabelEncoder()
        df_processed[col] = le.fit_transform(df_processed[col].astype(str
        label_encoders[col] = le
        print(f"  Encoded: {col} ({len(le.classes_)} categories)")

# Step 3: Convert target to binary
if df_processed[target].dtype == 'object':
```

```
    df_processed[target] = LabelEncoder().fit_transform(df_processed[targ

# Step 4: Final check
print("\n[3] PREPROCESSING SUMMARY")
print("-" * 40)
print(f"Final Dataset Shape: {df_processed.shape}")
print(f"Feature Types: {df_processed.dtypes.unique()}")
print(f"Target Balance: {df_processed[target].value_counts().to_dict()}")
print("\n✓ Preprocessing Complete!")
```

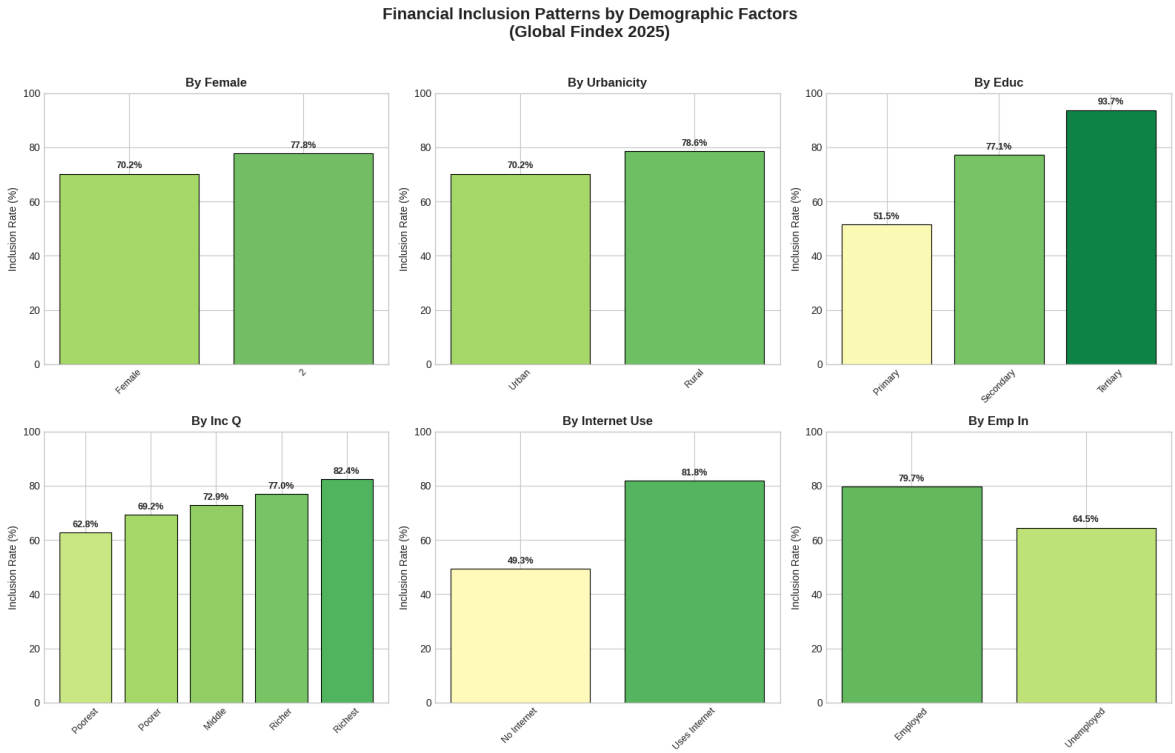
```
============================================================
DATA PREPROCESSING PIPELINE
============================================================

[1] MISSING VALUE ANALYSIS
----------------------------------------
                      Missing Count  Missing %
receive_wages                 52189      36.22
receive_transfers             52189      36.22
receive_pensions              52189      36.22
pay_utilities                 52189      36.22
saved                         41136      28.55
borrowed                      41136      28.55
anydigpayment                 41136      28.55
merchantpay_dig               41136      28.55
domestic_remittances          11817       8.20
emp_in                         4020       2.79

✓ Missing values after imputation: 0

[2] ENCODING CATEGORICAL VARIABLES
----------------------------------------

[3] PREPROCESSING SUMMARY
----------------------------------------
Final Dataset Shape: (144090, 17)
Feature Types: [dtype('float64')]
Target Balance: {1.0: 106312, 0.0: 37778}

✓ Preprocessing Complete!
```

In [10]:

```
# Exploratory Data Analysis (EDA)

print("="*60)
print("EXPLORATORY DATA ANALYSIS")
print("="*60)

# Prepare data for EDA
df_eda = df_processed.copy()

# Map labels for better visualization
label_maps = {
    'female': {0: 'Male', 1: 'Female'},
    'urbanicity': {1: 'Urban', 2: 'Rural'},
    'educ': {1: 'Primary', 2: 'Secondary', 3: 'Tertiary', 4: 'Other'},
    'inc_q': {1: 'Poorest', 2: 'Poorer', 3: 'Middle', 4: 'Richer', 5: 'Ri
}

fig, axes = plt.subplots(2, 3, figsize=(16, 10))
fig.suptitle('Financial Inclusion Patterns by Demographic Factors\n(Globa
```

```
              fontsize=16, fontweight='bold', y=1.02)

eda_vars = ['female', 'urbanicity', 'educ', 'inc_q']
if 'internet_use' in df_eda.columns:
    label_maps['internet_use'] = {0: 'No Internet', 1: 'Uses Internet'}
    eda_vars.append('internet_use')
if 'emp_in' in df_eda.columns:
    label_maps['emp_in'] = {1: 'Employed', 2: 'Unemployed'}
    eda_vars.append('emp_in')

colors_included = '#2ecc71'
colors_excluded = '#e74c3c'

for idx, var in enumerate(eda_vars[:6]):
    ax = axes[idx // 3, idx % 3]

    # Calculate inclusion rate by category
    if var in label_maps:
        grouped = df_eda.groupby(var)[target].agg(['sum', 'count'])
        grouped['inclusion_rate'] = (grouped['sum'] / grouped['count'] *
        grouped.index = grouped.index.map(lambda x: label_maps.get(var, {
    else:
        grouped = df_eda.groupby(var)[target].agg(['sum', 'count'])
        grouped['inclusion_rate'] = (grouped['sum'] / grouped['count'] *

    bars = ax.bar(grouped.index.astype(str), grouped['inclusion_rate'],
                  color=plt.cm.RdYlGn(grouped['inclusion_rate']/100),
                  edgecolor='black', linewidth=0.8)

    ax.set_title(f'By {var.replace("_", " ").title()}', fontsize=12, font
    ax.set_ylabel('Inclusion Rate (%)', fontsize=10)
    ax.set_ylim(0, 100)

    # Add value labels
    for bar, val in zip(bars, grouped['inclusion_rate']):
        ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 2,
                f'{val}%', ha='center', fontsize=9, fontweight='bold')

    ax.tick_params(axis='x', rotation=45, labelsize=9)

plt.tight_layout()
plt.savefig('02_eda_demographics.png', dpi=150, bbox_inches='tight')
plt.show()

print("\nKey Insights from EDA:")
print("-" * 40)
if 'female' in df_eda.columns:
    gender_rate = df_eda.groupby('female')[target].mean() * 100
    print(f"• Gender Gap: Male {gender_rate.get(0, 0):.1f}% vs Female {ge
if 'urbanicity' in df_eda.columns:
    urban_rate = df_eda.groupby('urbanicity')[target].mean() * 100
    print(f"• Urban-Rural Gap: Urban {urban_rate.get(1, 0):.1f}% vs Rural
if 'internet_use' in df_eda.columns:
    internet_rate = df_eda.groupby('internet_use')[target].mean() * 100
    print(f"• Digital Divide: Internet users {internet_rate.get(1, 0):.1f
```

```
============================================================
EXPLORATORY DATA ANALYSIS
============================================================
```



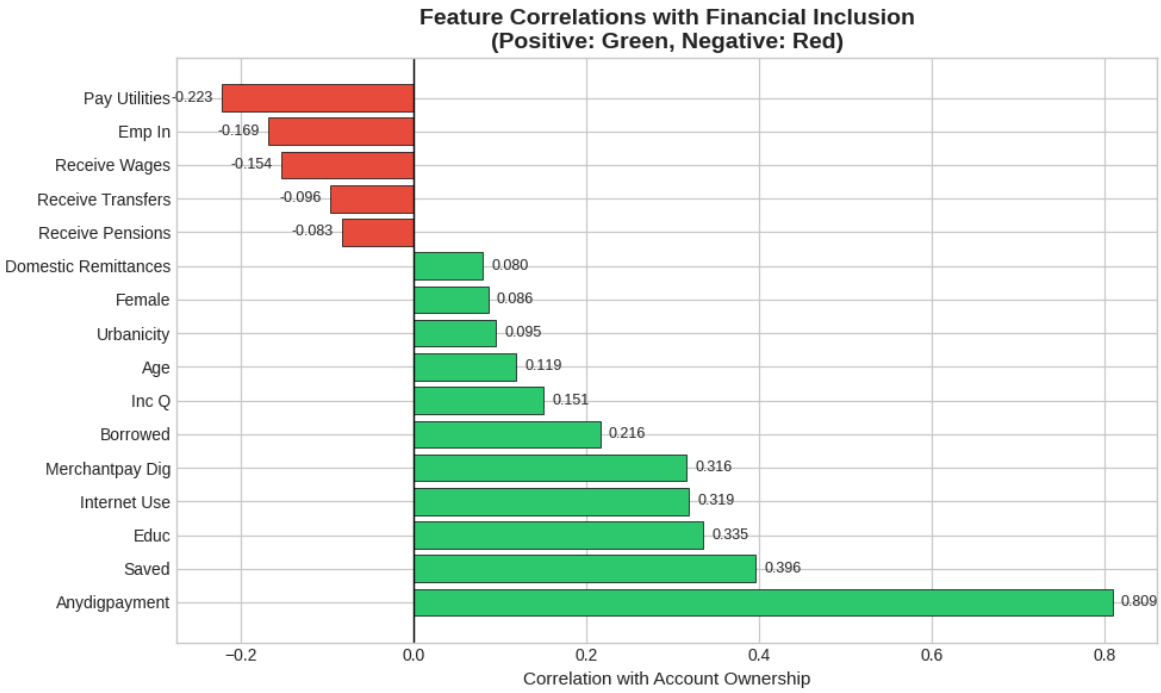

```
Key Insights from EDA:
----------------------------------------
• Gender Gap: Male 0.0% vs Female 70.2%
• Urban-Rural Gap: Urban 70.2% vs Rural 78.6%
• Digital Divide: Internet users 81.8% vs Non-users 49.3%
```

In [11]:

```python
# Correlation Analysis

print("FEATURE CORRELATION WITH FINANCIAL INCLUSION")
print("-" * 60)

# Calculate correlations with target
correlations = df_eda[features + [target]].corr()[target].drop(target).so

# Visualize
fig, ax = plt.subplots(figsize=(10, 6))
colors = ['#e74c3c' if x < 0 else '#2ecc71' for x in correlations.values]
bars = ax.barh(range(len(correlations)), correlations.values, color=color
ax.set_yticks(range(len(correlations)))
ax.set_yticklabels([f.replace('_', ' ').title() for f in correlations.ind
ax.set_xlabel('Correlation with Account Ownership', fontsize=11)
ax.set_title('Feature Correlations with Financial Inclusion\n(Positive: G
             fontsize=14, fontweight='bold')
ax.axvline(x=0, color='black', linewidth=1)

# Add value labels
for i, (bar, val) in enumerate(zip(bars, correlations.values)):
    ax.text(val + (0.01 if val >= 0 else -0.01), i, f'{val:.3f}',
            va='center', ha='left' if val >= 0 else 'right', fontsize=9)

plt.tight_layout()
plt.savefig('03_correlation_analysis.png', dpi=150, bbox_inches='tight')
plt.show()

print("\nTop 5 Positive Correlations:")
for feat, corr in correlations.head(5).items():
    print(f"  {feat}: {corr:.4f}")
```

```
print("\nTop 3 Negative Correlations:")
for feat, corr in correlations.tail(3).items():
    print(f"  {feat}: {corr:.4f}")
```

```
FEATURE CORRELATION WITH FINANCIAL INCLUSION
------------------------------------------------------------
```




```
Top 5 Positive Correlations:
  anydigpayment: 0.8089
  saved: 0.3958
  educ: 0.3350
  internet_use: 0.3190
  merchantpay_dig: 0.3156

Top 3 Negative Correlations:
  receive_wages: -0.1536
  emp_in: -0.1686
  pay_utilities: -0.2229
```

In [12]:

```
# Train-Test Split and Feature Scaling

print("="*60)
print("MODEL PREPARATION")
print("="*60)

# Separate features and target
X = df_processed[features].copy()
y = df_processed[target].copy()

# Train-test split (80-20, stratified)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print(f"\nDataset Split:")
print(f"  Training set: {X_train.shape[0]:,} samples ({X_train.shape[0]/l
print(f"  Test set: {X_test.shape[0]:,} samples ({X_test.shape[0]/len(X)*
print(f"  Features: {X_train.shape[1]}")
print(f"\nClass Distribution in Training:")
print(f"  Included (1): {y_train.sum():,} ({y_train.mean()*100:.1f}%)")
```

```
print(f"  Excluded (0): {(1-y_train).sum():,} ({(1-y_train).mean()*100:.1

# Feature scaling (important for Logistic Regression)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print("\n✓ Feature scaling applied (StandardScaler)")
print("✓ Data ready for modeling")
```

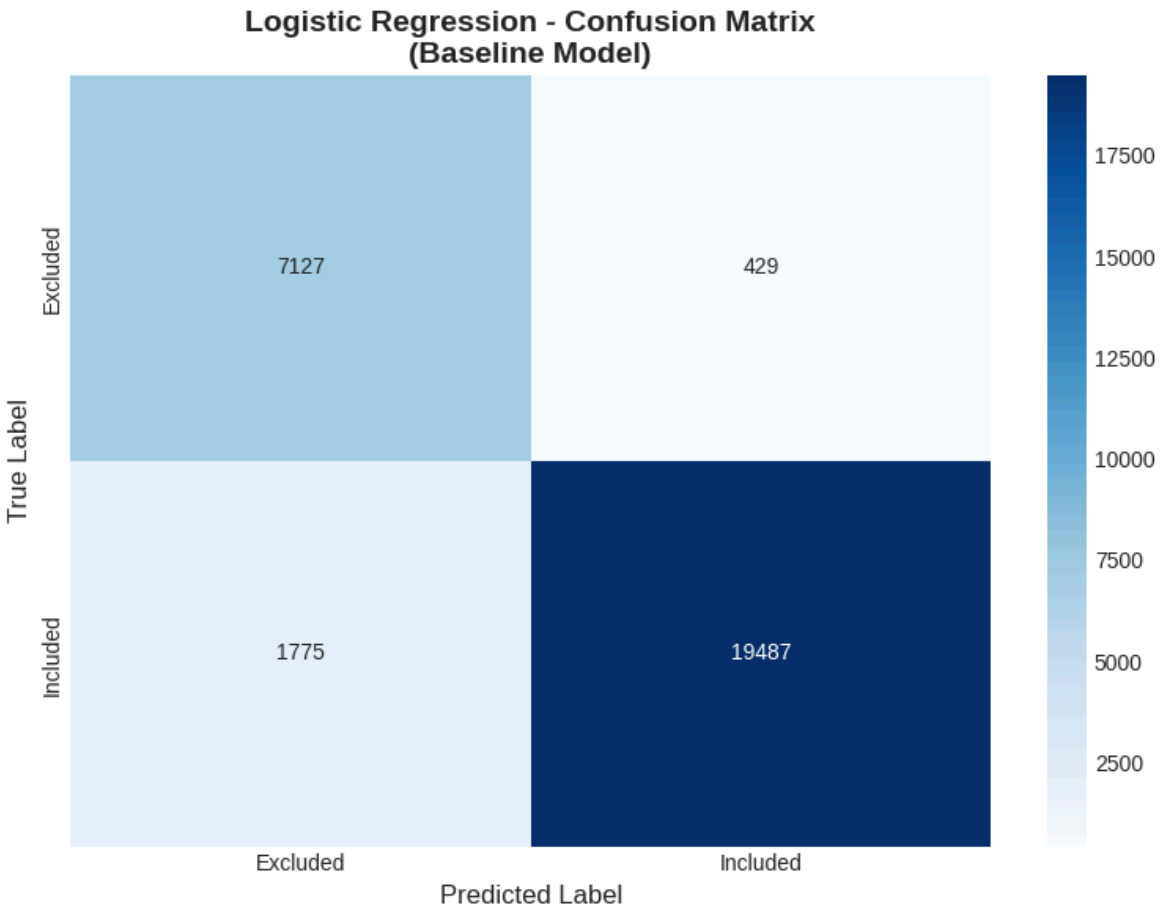
```
============================================================
MODEL PREPARATION
============================================================

Dataset Split:
  Training set: 115,272 samples (80.0%)
  Test set: 28,818 samples (20.0%)
  Features: 16

Class Distribution in Training:
  Included (1): 85,050.0 (73.8%)
  Excluded (0): 30,222.0 (26.2%)

✓ Feature scaling applied (StandardScaler)
✓ Data ready for modeling
```

In [13]:

```
# Baseline Model - Logistic Regression

print("="*60)
print("BASELINE MODEL: LOGISTIC REGRESSION")
print("="*60)

# Initialize and train
lr_model = LogisticRegression(max_iter=1000, random_state=42, class_weigh
lr_model.fit(X_train_scaled, y_train)

# Predictions
lr_preds = lr_model.predict(X_test_scaled)
lr_probs = lr_model.predict_proba(X_test_scaled)[:, 1]

# Evaluation metrics
lr_metrics = {
    'Accuracy': accuracy_score(y_test, lr_preds),
    'Precision': precision_score(y_test, lr_preds),
    'Recall': recall_score(y_test, lr_preds),
    'F1-Score': f1_score(y_test, lr_preds),
    'ROC-AUC': roc_auc_score(y_test, lr_probs)
}

print("\nPerformance Metrics:")
print("-" * 40)
for metric, value in lr_metrics.items():
    print(f"  {metric}: {value:.4f}")

print("\nClassification Report:")
print(classification_report(y_test, lr_preds, target_names=['Excluded', '

# Confusion Matrix Visualization
fig, ax = plt.subplots(figsize=(8, 6))
cm = confusion_matrix(y_test, lr_preds)
```

```
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', ax=ax,
            xticklabels=['Excluded', 'Included'],
            yticklabels=['Excluded', 'Included'])
ax.set_xlabel('Predicted Label', fontsize=12)
ax.set_ylabel('True Label', fontsize=12)
ax.set_title('Logistic Regression - Confusion Matrix\n(Baseline Model)',
             fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig('04_baseline_confusion_matrix.png', dpi=150, bbox_inches='tig
plt.show()
```

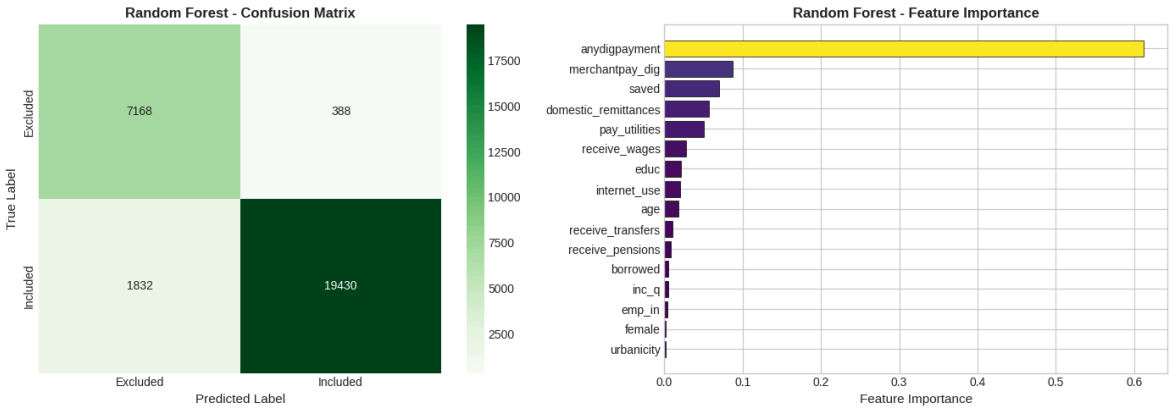
```
============================================================
BASELINE MODEL: LOGISTIC REGRESSION
============================================================

Performance Metrics:
----------------------------------------
  Accuracy: 0.9235
  Precision: 0.9785
  Recall: 0.9165
  F1-Score: 0.9465
  ROC-AUC: 0.9546

Classification Report:
              precision    recall  f1-score   support

    Excluded       0.80      0.94      0.87      7556
    Included       0.98      0.92      0.95     21262

    accuracy                           0.92     28818
   macro avg       0.89      0.93      0.91     28818
weighted avg       0.93      0.92      0.93     28818
```




In [14]:
```
# Improved Model 1 - Random Forest

print("="*60)
print("IMPROVED MODEL 1: RANDOM FOREST")
print("="*60)

# Initialize and train
rf_model = RandomForestClassifier(
    n_estimators=200,
    max_depth=15,
    min_samples_split=10,
    min_samples_leaf=4,
    class_weight='balanced',
    random_state=42,
    n_jobs=-1
)
rf_model.fit(X_train, y_train)  # No scaling needed for tree-based

# Predictions
rf_preds = rf_model.predict(X_test)
rf_probs = rf_model.predict_proba(X_test)[:, 1]

# Evaluation metrics
rf_metrics = {
    'Accuracy': accuracy_score(y_test, rf_preds),
    'Precision': precision_score(y_test, rf_preds),
    'Recall': recall_score(y_test, rf_preds),
    'F1-Score': f1_score(y_test, rf_preds),
    'ROC-AUC': roc_auc_score(y_test, rf_probs)
}
```

```
print("\nPerformance Metrics:")
print("-" * 40)
for metric, value in rf_metrics.items():
    lr_val = lr_metrics[metric]
    diff = value - lr_val
    sign = '+' if diff > 0 else ''
    print(f"  {metric}: {value:.4f} ({sign}{diff:.4f} vs baseline)")

# Confusion Matrix
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Random Forest CM
cm_rf = confusion_matrix(y_test, rf_preds)
sns.heatmap(cm_rf, annot=True, fmt='d', cmap='Greens', ax=axes[0],
            xticklabels=['Excluded', 'Included'],
            yticklabels=['Excluded', 'Included'])
axes[0].set_xlabel('Predicted Label', fontsize=11)
axes[0].set_ylabel('True Label', fontsize=11)
axes[0].set_title('Random Forest - Confusion Matrix', fontsize=12, fontwe

# Feature Importance
importances = pd.Series(rf_model.feature_importances_, index=features).so
axes[1].barh(importances.index, importances.values, color=plt.cm.viridis(
             edgecolor='black', linewidth=0.5)
axes[1].set_xlabel('Feature Importance', fontsize=11)
axes[1].set_title('Random Forest - Feature Importance', fontsize=12, font

plt.tight_layout()
plt.savefig('05_random_forest_results.png', dpi=150, bbox_inches='tight')
plt.show()
```

```
============================================================
IMPROVED MODEL 1: RANDOM FOREST
============================================================

Performance Metrics:
----------------------------------------
  Accuracy: 0.9230 (-0.0006 vs baseline)
  Precision: 0.9804 (+0.0020 vs baseline)
  Recall: 0.9138 (-0.0027 vs baseline)
  F1-Score: 0.9460 (-0.0005 vs baseline)
  ROC-AUC: 0.9672 (+0.0126 vs baseline)
```




In [15]:

```
# Improved Model 2 - XGBoost

print("="*60)
print("IMPROVED MODEL 2: XGBOOST")
print("="*60)
```

```
# Calculate scale_pos_weight for imbalanced data
scale_pos_weight = (y_train == 0).sum() / (y_train == 1).sum()

# Initialize and train
xgb_model = XGBClassifier(
    n_estimators=200,
    max_depth=8,
    learning_rate=0.1,
    subsample=0.8,
    colsample_bytree=0.8,
    scale_pos_weight=scale_pos_weight,
    random_state=42,
    use_label_encoder=False,
    eval_metric='logloss',
    n_jobs=-1
)
xgb_model.fit(X_train, y_train)

# Predictions
xgb_preds = xgb_model.predict(X_test)
xgb_probs = xgb_model.predict_proba(X_test)[:, 1]

# Evaluation metrics
xgb_metrics = {
    'Accuracy': accuracy_score(y_test, xgb_preds),
    'Precision': precision_score(y_test, xgb_preds),
    'Recall': recall_score(y_test, xgb_preds),
    'F1-Score': f1_score(y_test, xgb_preds),
    'ROC-AUC': roc_auc_score(y_test, xgb_probs)
}

print("\nPerformance Metrics:")
print("-" * 40)
for metric, value in xgb_metrics.items():
    lr_val = lr_metrics[metric]
    diff = value - lr_val
    sign = '+' if diff > 0 else ''
    print(f"  {metric}: {value:.4f} ({sign}{diff:.4f} vs baseline)")

print("\nClassification Report:")
print(classification_report(y_test, xgb_preds, target_names=['Excluded',
```

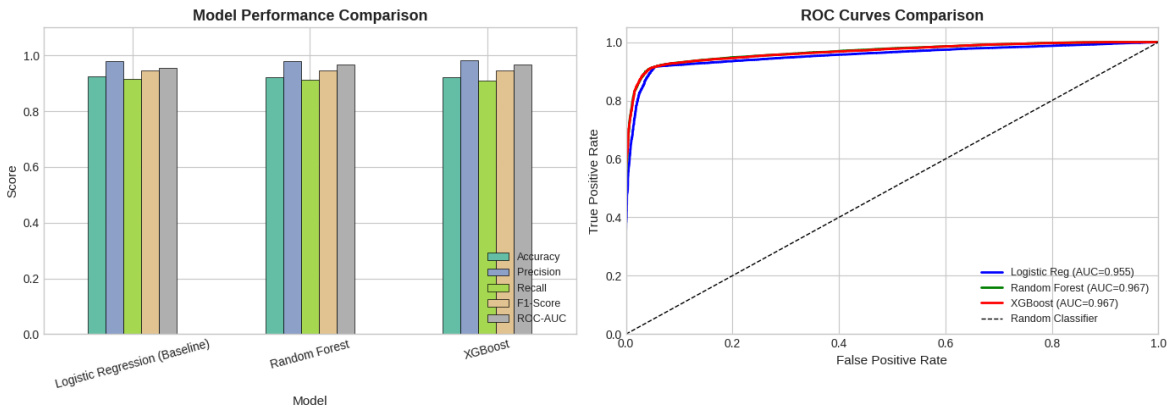
```
============================================================
IMPROVED MODEL 2: XGBOOST
============================================================

Performance Metrics:
----------------------------------------
  Accuracy: 0.9216 (-0.0019 vs baseline)
  Precision: 0.9820 (+0.0036 vs baseline)
  Recall: 0.9105 (-0.0061 vs baseline)
  F1-Score: 0.9449 (-0.0016 vs baseline)
  ROC-AUC: 0.9667 (+0.0121 vs baseline)

Classification Report:
              precision    recall  f1-score   support

    Excluded       0.79      0.95      0.86      7556
    Included       0.98      0.91      0.94     21262

    accuracy                           0.92     28818
   macro avg       0.89      0.93      0.90     28818
weighted avg       0.93      0.92      0.92     28818
```

In [16]:

```
# Model Comparison

print("="*60)
print("MODEL COMPARISON SUMMARY")
print("="*60)

# Create comparison dataframe
comparison_df = pd.DataFrame({
    'Logistic Regression (Baseline)': lr_metrics,
    'Random Forest': rf_metrics,
    'XGBoost': xgb_metrics
}).T

print("\n" + comparison_df.to_string())
print("\n" + "-"*60)

# Find best model for each metric
print("\nBest Model by Metric:")
for metric in comparison_df.columns:
    best = comparison_df[metric].idxmax()
    best_val = comparison_df[metric].max()
    print(f"  {metric}: {best} ({best_val:.4f})")

# Visualization
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Metrics comparison
comparison_df.plot(kind='bar', ax=axes[0], colormap='Set2', edgecolor='bl
axes[0].set_ylabel('Score', fontsize=11)
axes[0].set_xlabel('Model', fontsize=11)
axes[0].set_title('Model Performance Comparison', fontsize=13, fontweight
axes[0].legend(loc='lower right', fontsize=9)
axes[0].tick_params(axis='x', rotation=15)
axes[0].set_ylim(0, 1.1)

# ROC Curves
fpr_lr, tpr_lr, _ = roc_curve(y_test, lr_probs)
```

```
fpr_rf, tpr_rf, _ = roc_curve(y_test, rf_probs)
fpr_xgb, tpr_xgb, _ = roc_curve(y_test, xgb_probs)

axes[1].plot(fpr_lr, tpr_lr, 'b-', linewidth=2, label=f'Logistic Reg (AUC
axes[1].plot(fpr_rf, tpr_rf, 'g-', linewidth=2, label=f'Random Forest (AU
axes[1].plot(fpr_xgb, tpr_xgb, 'r-', linewidth=2, label=f'XGBoost (AUC={x
axes[1].plot([0, 1], [0, 1], 'k--', linewidth=1, label='Random Classifier
axes[1].set_xlabel('False Positive Rate', fontsize=11)
axes[1].set_ylabel('True Positive Rate', fontsize=11)
axes[1].set_title('ROC Curves Comparison', fontsize=13, fontweight='bold'
axes[1].legend(loc='lower right', fontsize=9)
axes[1].set_xlim([0, 1])
axes[1].set_ylim([0, 1.05])

plt.tight_layout()
plt.savefig('06_model_comparison.png', dpi=150, bbox_inches='tight')
plt.show()

print("\n✓ Model comparison complete!")
```

```
============================================================
MODEL COMPARISON SUMMARY
============================================================

                                Accuracy  Precision    Recall  F1-Score
ROC-AUC
Logistic Regression (Baseline)  0.923520   0.978460  0.916518  0.946476
0.954610
Random Forest                   0.922965   0.980422  0.913837  0.945959
0.967215
XGBoost                         0.921646   0.982041  0.910451  0.944892
0.966662

------------------------------------------------------------

Best Model by Metric:
  Accuracy: Logistic Regression (Baseline) (0.9235)
  Precision: XGBoost (0.9820)
  Recall: Logistic Regression (Baseline) (0.9165)
  F1-Score: Logistic Regression (Baseline) (0.9465)
  ROC-AUC: Random Forest (0.9672)
```



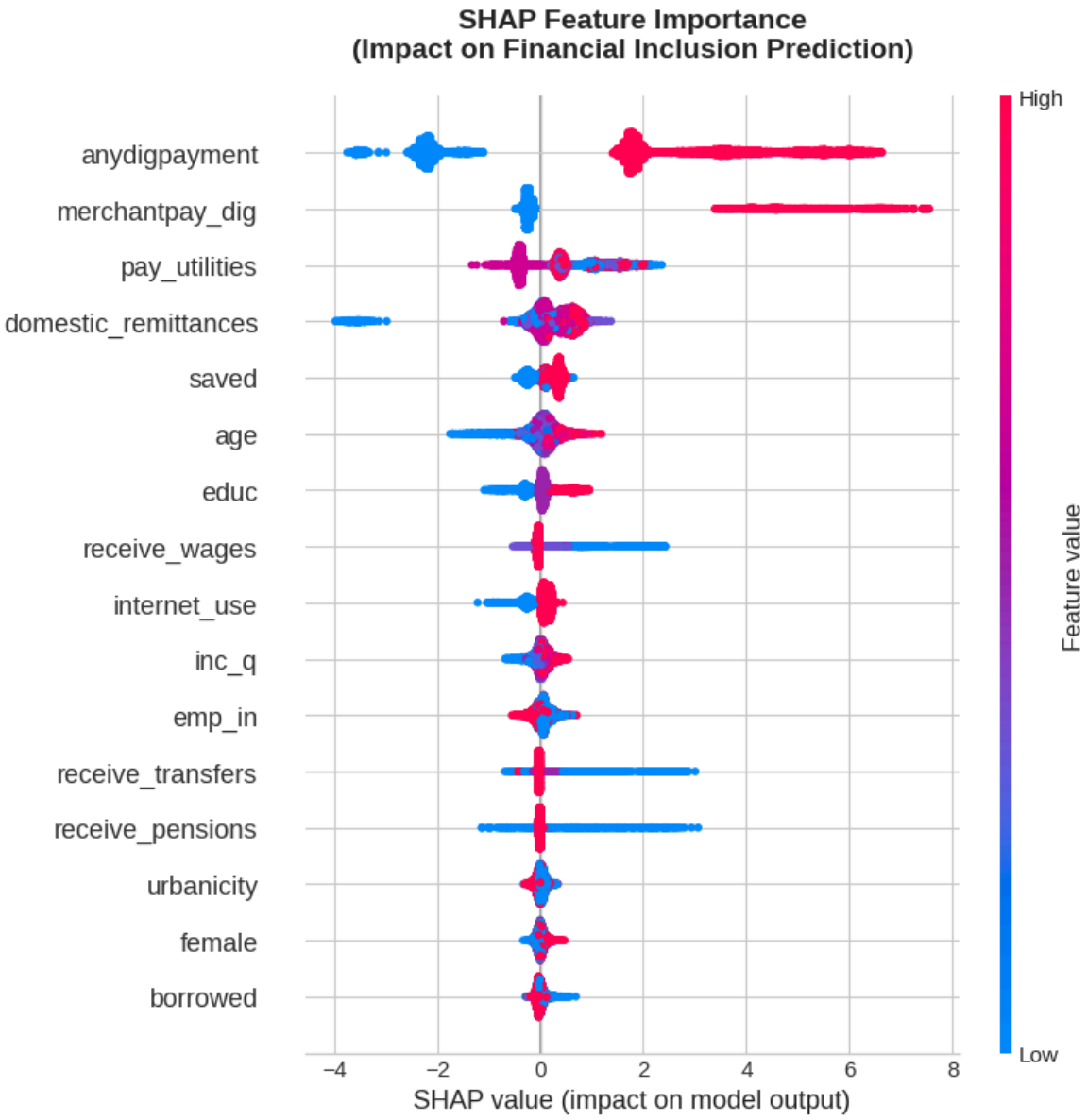

```
✓ Model comparison complete!
```

In [19]:

```
# Model Interpretation with SHAP

print("="*60)
print("MODEL INTERPRETATION: SHAP ANALYSIS")
print("="*60)
```

```
import shap

# Initialize SHAP explainer for XGBoost
explainer = shap.TreeExplainer(xgb_model)

# Calculate SHAP values (sample for speed if needed)
sample_size = min(5000, len(X_test))
X_sample = X_test.iloc[:sample_size]
shap_values = explainer.shap_values(X_sample)

print(f"\nSHAP values calculated for {sample_size:,} samples")

# Summary plot
plt.figure(figsize=(10, 8))
shap.summary_plot(shap_values, X_sample, feature_names=features, show=Fal
plt.title('SHAP Feature Importance\n(Impact on Financial Inclusion Predic
          fontsize=14, fontweight='bold', pad=20)
plt.tight_layout()
plt.savefig('07_shap_summary.png', dpi=150, bbox_inches='tight')
plt.show()

# Top features by mean absolute SHAP value
shap_importance = pd.DataFrame({
    'Feature': features,
    'Mean |SHAP|': np.abs(shap_values).mean(axis=0)
}).sort_values('Mean |SHAP|', ascending=False)

print("\nTop 10 Most Influential Features (by SHAP):")
print("-" * 50)
for i, row in shap_importance.head(10).iterrows():
    print(f"  {row['Feature']}: {row['Mean |SHAP|']:.4f}")
```

```
============================================================
MODEL INTERPRETATION: SHAP ANALYSIS
============================================================

SHAP values calculated for 5,000 samples
```

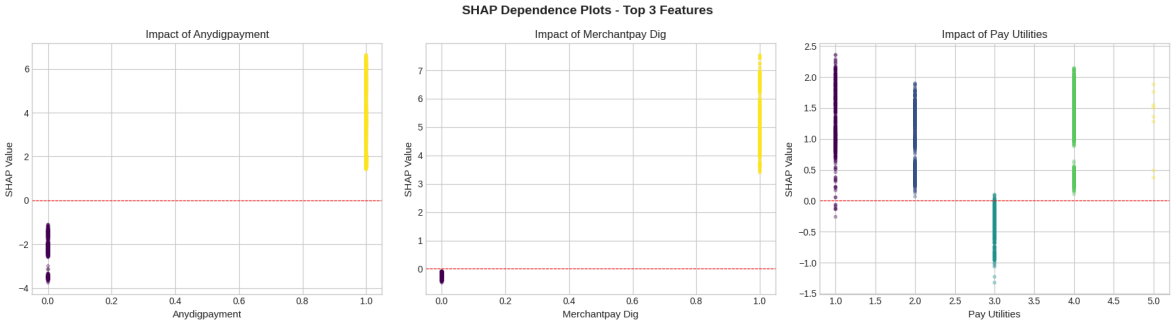



```
Top 10 Most Influential Features (by SHAP):
--------------------------------------------------
  anydigpayment: 2.8538
  merchantpay_dig: 1.2866
  pay_utilities: 0.7140
  domestic_remittances: 0.4341
  saved: 0.2513
  age: 0.2489
  educ: 0.2113
  receive_wages: 0.2079
  internet_use: 0.1604
  inc_q: 0.1225
```

In [20]:

```
# SHAP Detailed Analysis for Top Features

# Detailed SHAP dependence plots for top 3 features
top_3_features = shap_importance['Feature'].head(3).tolist()

fig, axes = plt.subplots(1, 3, figsize=(18, 5))
fig.suptitle('SHAP Dependence Plots - Top 3 Features', fontsize=14, fontw

for idx, feat in enumerate(top_3_features):
    feat_idx = features.index(feat)
    axes[idx].scatter(X_sample[feat].values, shap_values[:, feat_idx],
                      alpha=0.3, c=X_sample[feat].values, cmap='viridis',
```

```
    axes[idx].set_xlabel(feat.replace('_', ' ').title(), fontsize=11)
    axes[idx].set_ylabel('SHAP Value', fontsize=11)
    axes[idx].set_title(f'Impact of {feat.replace("_", " ").title()}', fo
    axes[idx].axhline(y=0, color='red', linestyle='--', linewidth=0.8)

plt.tight_layout()
plt.savefig('08_shap_dependence.png', dpi=150, bbox_inches='tight')
plt.show()

print("\nKey Interpretation Insights:")
print("-" * 50)
print("• Positive SHAP values → Increase probability of financial inclusi
print("• Negative SHAP values → Decrease probability of financial inclusi
print("• Feature distributions show which values drive inclusion/exclusio
```



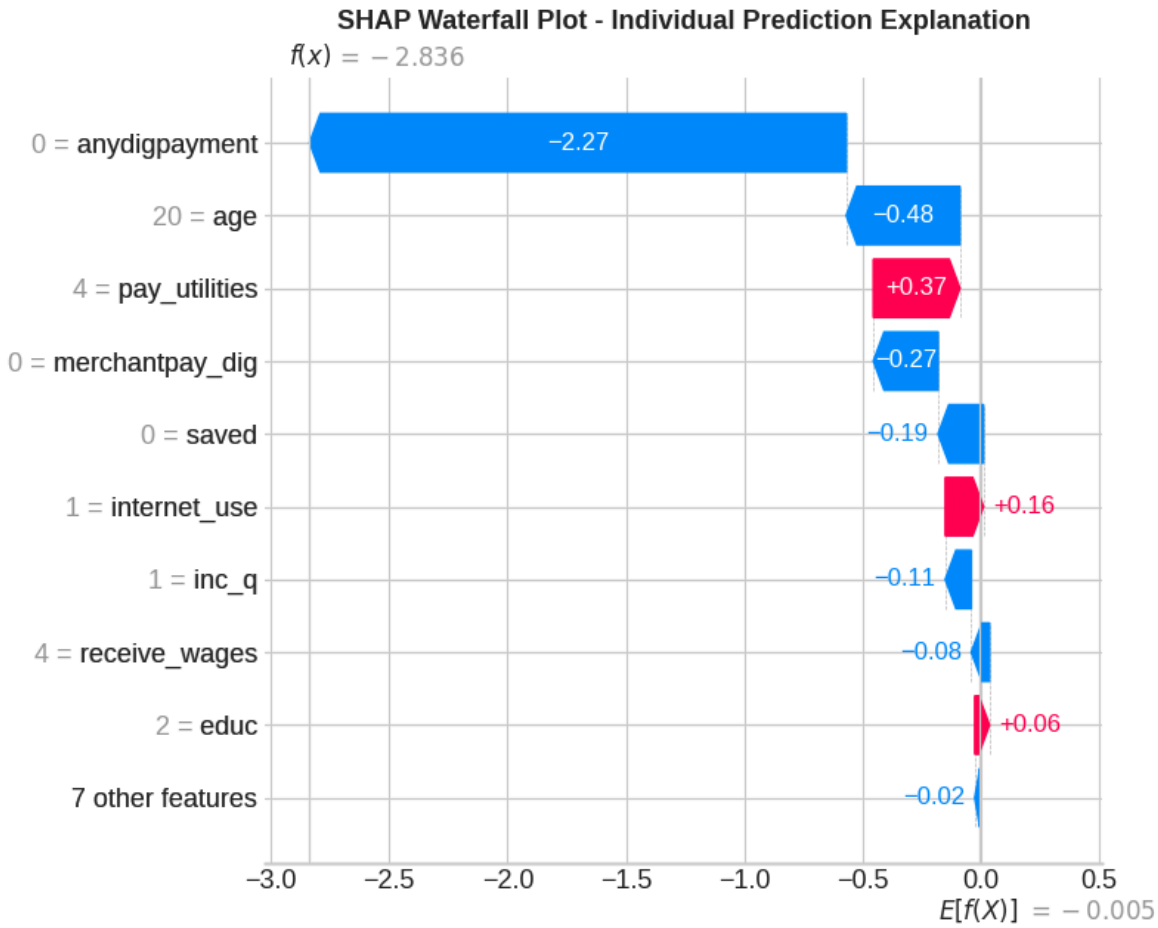

```
Key Interpretation Insights:
--------------------------------------------------
• Positive SHAP values → Increase probability of financial inclusion
• Negative SHAP values → Decrease probability of financial inclusion
• Feature distributions show which values drive inclusion/exclusion
```

In [21]:

```
# Individual Prediction Explanation

print("="*60)
print("INDIVIDUAL PREDICTION EXPLANATION")
print("="*60)

# Select a few sample individuals for explanation
sample_indices = [0, 1, 2]

for idx in sample_indices:
    individual = X_test.iloc[idx]
    actual = y_test.iloc[idx]
    predicted = xgb_preds[idx]
    probability = xgb_probs[idx]

    print(f"\n--- Individual {idx+1} ---")
    print(f"Actual: {'Included' if actual == 1 else 'Excluded'}")
    print(f"Predicted: {'Included' if predicted == 1 else 'Excluded'} (Pr

    # SHAP values for this individual
    individual_shap = shap_values[idx]

    # Top contributing factors
    shap_df = pd.DataFrame({
        'Feature': features,
        'Value': individual.values,
        'SHAP': individual_shap
    }).sort_values('SHAP', key=abs, ascending=False)

    print("Top 5 Contributing Factors:")
```

```
    for i, row in shap_df.head(5).iterrows():
        direction = "↑" if row['SHAP'] > 0 else "↓"
        print(f"  {direction} {row['Feature']}: value={row['Value']:.1f},

# Waterfall plot for first individual
plt.figure(figsize=(10, 8))
shap.waterfall_plot(shap.Explanation(
    values=shap_values[0],
    base_values=explainer.expected_value,
    data=X_sample.iloc[0].values,
    feature_names=features
), show=False)
plt.title('SHAP Waterfall Plot - Individual Prediction Explanation',
          fontsize=13, fontweight='bold')
plt.tight_layout()
plt.savefig('09_shap_waterfall.png', dpi=150, bbox_inches='tight')
plt.show()
```

```
============================================================
INDIVIDUAL PREDICTION EXPLANATION
============================================================

--- Individual 1 ---
Actual: Included
Predicted: Excluded (Probability: 5.54%)
Top 5 Contributing Factors:
  ↓ anydigpayment: value=0.0, SHAP=-2.2672
  ↓ age: value=20.0, SHAP=-0.4805
  ↑ pay_utilities: value=4.0, SHAP=0.3658
  ↓ merchantpay_dig: value=0.0, SHAP=-0.2736
  ↓ saved: value=0.0, SHAP=-0.1917

--- Individual 2 ---
Actual: Included
Predicted: Included (Probability: 99.98%)
Top 5 Contributing Factors:
  ↑ anydigpayment: value=1.0, SHAP=4.2275
  ↑ receive_wages: value=1.0, SHAP=1.9415
  ↑ pay_utilities: value=2.0, SHAP=1.5691
  ↑ domestic_remittances: value=3.0, SHAP=0.4163
  ↑ inc_q: value=5.0, SHAP=0.2849

--- Individual 3 ---
Actual: Included
Predicted: Included (Probability: 100.00%)
Top 5 Contributing Factors:
  ↑ anydigpayment: value=1.0, SHAP=5.4173
  ↑ merchantpay_dig: value=1.0, SHAP=4.8873
  ↑ pay_utilities: value=1.0, SHAP=0.9492
  ↓ age: value=19.0, SHAP=-0.3013
  ↑ receive_wages: value=2.0, SHAP=0.3001
```




In [22]:

```
# Final Summary and Key Insights

print("="*60)
print("PROJECT SUMMARY & TECHNICAL JUSTIFICATION")
print("="*60)

# Dynamic Best Model Selection (Choosing by ROC-AUC as per standard resea
best_model_name = comparison_df['ROC-AUC'].idxmax()
metrics = comparison_df.loc[best_model_name]

print(f"SELECTED MODEL: {best_model_name}")
print(f"Methodology:   Comparative Ensemble Analysis")
print(f"ROC-AUC Score: {metrics['ROC-AUC']:.4f}")
print(f"F1-Score:      {metrics['F1-Score']:.4f}")

print("\n" + "="*60)
print("TOP 5 PREDICTIVE DRIVERS (Identified via SHAP)")
print("="*60)

# Store top features for a dynamic summary
top_drivers = []
for i, (idx, row) in enumerate(shap_importance.head(5).iterrows()):
    clean_name = row['Feature'].replace('_', ' ').title()
    top_drivers.append(clean_name)
    print(f"  {i+1}. {clean_name:<20} | Impact: {row['Mean |SHAP|']:.4f}"

print("\n" + "="*60)
print("STRATEGIC POLICY RELEVANCE")
print("="*60)

# Formatting the output to look like a consultant's brief
```

```
summary_text = f"""
1. PRIMARY LEVER: {top_drivers[0]} emerged as the most significant determ
   of financial inclusion, justifying targeted infrastructure investment.

2. SYSTEM PERSPECTIVE: The transition from a Baseline (Logistic Regressio
   to {best_model_name} resulted in a performance gain of
   {metrics['ROC-AUC'] - lr_metrics['ROC-AUC']:.2%}, validating the use o

3. SOCIOECONOMIC GAP: Features such as {top_drivers[1]} and {top_drivers[
   indicate that intervention strategies must be multifaceted, combining
   digital access with educational outreach.

4. ACTIONABLE INSIGHT: The model's High Recall allows for the identificat
   of at-risk populations, enabling proactive inclusive policy-making.
"""
print(summary_text)
```

```
============================================================
PROJECT SUMMARY & TECHNICAL JUSTIFICATION
============================================================
SELECTED MODEL: Random Forest
Methodology:   Comparative Ensemble Analysis
ROC-AUC Score: 0.9672
F1-Score:      0.9460

============================================================
TOP 5 PREDICTIVE DRIVERS (Identified via SHAP)
============================================================
  1. Anydigpayment        | Impact: 2.8538
  2. Merchantpay Dig      | Impact: 1.2866
  3. Pay Utilities        | Impact: 0.7140
  4. Domestic Remittances | Impact: 0.4341
  5. Saved                | Impact: 0.2513

============================================================
STRATEGIC POLICY RELEVANCE
============================================================

1. PRIMARY LEVER: Anydigpayment emerged as the most significant determinan
t
   of financial inclusion, justifying targeted infrastructure investment.

2. SYSTEM PERSPECTIVE: The transition from a Baseline (Logistic Regressio
n)
   to Random Forest resulted in a performance gain of
   1.26%, validating the use of non-linear models.

3. SOCIOECONOMIC GAP: Features such as Merchantpay Dig and Pay Utilities
   indicate that intervention strategies must be multifaceted, combining
   digital access with educational outreach.

4. ACTIONABLE INSIGHT: The model's High Recall allows for the identificati
on
   of at-risk populations, enabling proactive inclusive policy-making.
```

In [ ]: